# NATHANIEL POWELL

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## EDUCATION

**University of Waterloo** Waterloo, ON

*Bachelor of Science, Honours Mathematical Physics, Co-op* Sep. 2025 - Present

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
**Web & Data:** React, Vite, Node.js, Express, Next.js, FastAPI, PostgreSQL, Supabase, REST APIs, WebSockets
**AI & Simulation:** PyTorch, Stable-Baselines3, PPO, OpenCV, SUMO, TraCI, HOG/SVM
**Engineering:** Docker, Git, GitHub Actions, CI/CD, API testing, authentication, data modeling, Figma

## EXPERIENCE

**Quadmates.ca** 2026 - Present

*Founder & Full-Stack Developer*

- Shipped a full-stack student collaboration platform using React, Node.js/Express, PostgreSQL and Supabase with verified accounts, partner matching, group messaging and study-time tracking.
- Reduced direct/group messaging latency by 70% (internally reported); merged live chat events with history and moved read receipts to atomic SQL updates to avoid concurrent-write conflicts.
- Developed privacy-first analytics with anonymous session tracking, public page views and Do Not Track controls; collected 50 views across 27 sessions in the initial November 2026 rollout.
- Built an opt-in AI study coach/planner with input validation and server-side usage quotas; prepared an iOS beta foundation in Capacitor that reuses existing backend services.

**Infosys InStep** Jan - Apr 2027

*Incoming Computer Engineering Assistant - AI Agent Engineering*

- Selected for Infosys's CTO office in Bengaluru, India, for an upcoming placement focused on AI agents and enterprise process automation.

**UW Orbital** 2026 - Present

*Business & Finance Team Member*

- Negotiated corporate manufacturing and resource support for CubeSat engineering, resolving sponsor production-schedule constraints through technical documentation and compromise.
- Coordinated technical stakeholders and supported engineering budgets using Excel and Python automation prototypes.

## SELECTED ENGINEERING PROJECTS

**TransitOps** | *Python, SUMO/TraCI, FastAPI, Next.js, Docker*

- Built a deterministic transit-control simulator modeling three buses and three signalized intersections, streaming SUMO telemetry to a web dashboard and validating integration with CI smoke tests.

**NeuroSim 2.0** | *Python, PyTorch, PPO, Stable-Baselines3*

- Developed a 2D reinforcement-learning rocket-landing environment and evaluated policies by landing success, touchdown error, crashes, and fuel consumption.

**Computer Vision Security Camera** | *Python, OpenCV, Tkinter*

- Created real-time person detection using HOG/SVM, a multithreaded producer-consumer pipeline, bounded buffering and frame sampling.